Calculation of genotype values (BLUEs) emmeans a pisum

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Setup

Packages

```
suppressPackageStartupMessages(library("tidyverse"))
suppressPackageStartupMessages(library("lme4"))
suppressPackageStartupMessages(library("Hmisc"))
suppressPackageStartupMessages(library("corrplot"))
suppressPackageStartupMessages(library("emmeans"))
suppressPackageStartupMessages(library("performance"))
suppressPackageStartupMessages(library("MASS"))
```

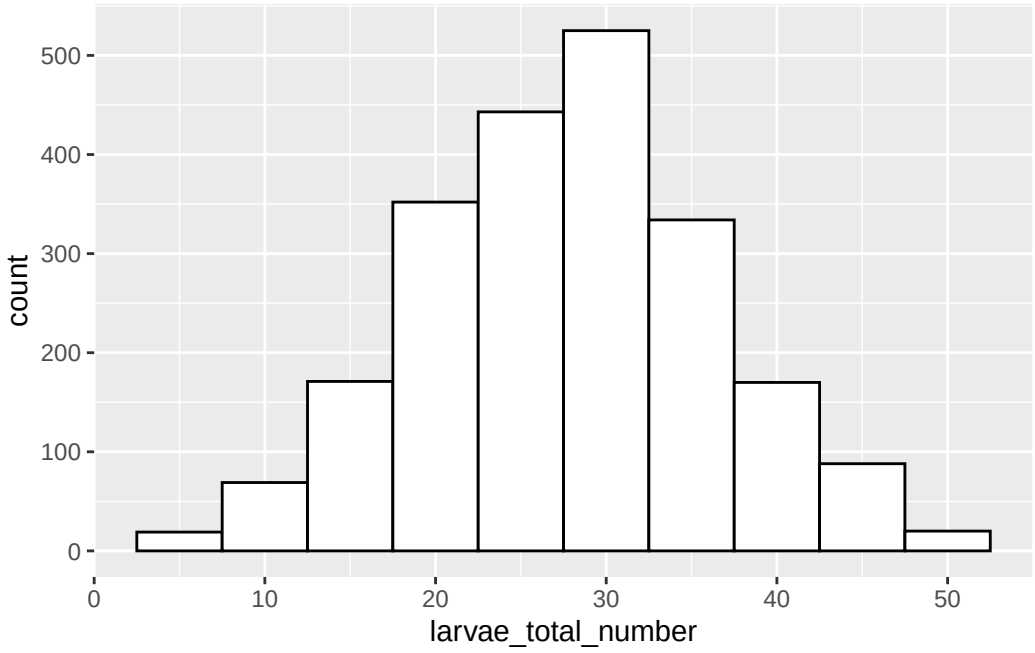
Filtered data import

```
data250pisum <- read.csv("../../01.dataset_filtering/02_A.pisum/final filters/acyrthosiphon_p
data250pisum$genotype <- as.factor(data250pisum$ivip)
```

analysis are done on only one dataset with minimal number of rep equal to 5 but the same finding are observed on dataset with other nimimal number of rep

Data distribution

```
data250pisum %>%
  ggplot(., aes(x = larvae_total_number)) +
  geom_histogram(binwidth = 5, fill = "white", color = "black")
```



Linear model fitting and emmeans extraction

Useful lesson on LMMs.

Fit linear models

We tested few model to see which one is better to estimate the larvae fecundity

```
model0 <- lm(larvae_total_number ~ 1, data = data250pisum)
model1 <- lm(larvae_total_number ~ ivip, data = data250pisum)
model2 <- lmer(larvae_total_number ~ ivip + (1 | session), data = data250pisum)
model3 <- lmer(larvae_total_number ~ ivip + (1 | session/block), data = data250pisum)
model4 <- glm.nb(formula = larvae_total_number ~ ivip +session/block, data = data250pisum)
model5 <- glm.nb(formula = larvae_total_number ~ ivip +session, data = data250pisum)
```

```
compare_performance(model0, model1, model2,model3,model4,model5, rank = FALSE, verbose = TRU
```

```
When comparing models, please note that probably not all models were fit
  from same data.

# Comparison of Model Performance Indices

Name   |   Model |   AIC (weights) |  AICc (weights) |   BIC (weights) |  RMSE
-------------------------------------------------------------------------------
model0 |      lm | 15636.1 (<.001) | 15636.1 (<.001) | 15647.4 (>.999) | 8.571
model1 |      lm | 15449.4 (<.001) | 15500.1 (<.001) | 16703.9 (<.001) | 7.845
model2 | lmerMod | 15249.9 (<.001) | 15301.1 (<.001) | 16510.1 (<.001) | 7.429
model3 | lmerMod | 15059.5 (>.999) | 15111.2 (>.999) | 16325.4 (<.001) | 6.682
model4 |  negbin | 15114.8 (<.001) | 15229.0 (<.001) | 16948.3 (<.001) | 6.658
model5 |  negbin | 15374.8 (<.001) | 15428.4 (<.001) | 16663.3 (<.001) | 7.429

Name   | Sigma | Nagelkerke's R2 | Score_log | Score_spherical | R2 (cond.)
-------------------------------------------------------------------------
model0 | 8.573 |                 |           |                 |
model1 | 8.278 |                 |           |                 |
model2 | 7.852 |                 |           |                 |      0.258
model3 | 7.203 |                 |           |                 |      0.395
model4 | 1.000 |           0.570 |    -3.796 |           0.021 |
model5 | 1.000 |           0.359 |    -3.850 |           0.021 |

Name   | R2 (marg.) |   ICC |    R2 | R2 (adj.)
-----------------------------------------------
model0 |            |       | 0.000 |     0.000
model1 |            |       | 0.165 |     0.071
model2 |      0.150 | 0.128 |       |
```

```
model3 |      0.146 | 0.291 |       |
model4 |            |       |       |
model5 |            |       |       |
```

> **!** Important
>
> Model3 that models session as a random fixed factor and block nested within session is a slightly but significant better model.

Diagnostics

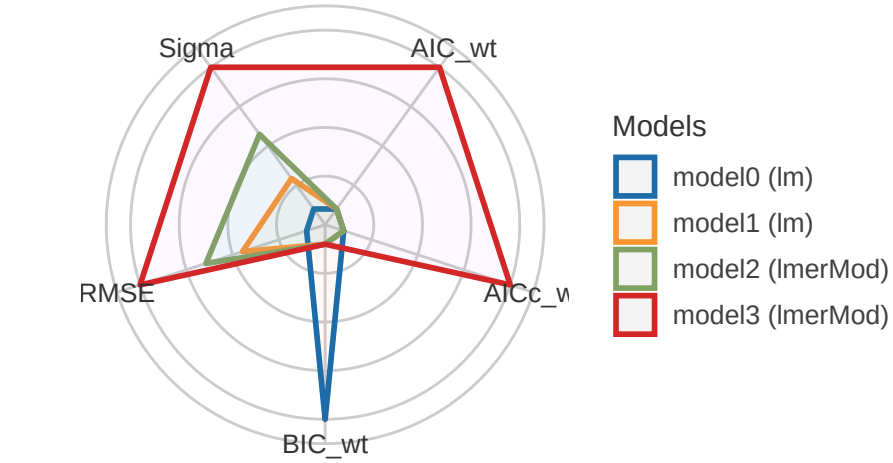
```
check_model(model3)
```

> **!** Important
>
> Most assumptions seem to be valid: normality of residuals, normality of random effects, linear relationship, homogeneity of variance, multicollinearity.

Model performances

```
plot(compare_performance(model0, model1, model2,model3))
```

```
When comparing models, please note that probably not all models were fit
  from same data.
```




```
test_lrt(model1, model2)
```

```
# Likelihood-Ratio-Test (LRT) for Model Comparison (ML-estimator)

Name   |   Model |  df | df_diff |   Chi2 |      p
-------------------------------------------------
model1 |      lm | 221 |         |        |
model2 | lmerMod | 222 |       1 | 201.42 | < .001
```

```
test_lrt(model2, model3)
```

```
Some of the nested models seem to be identical and probably only vary in
  their random effects.

# Likelihood-Ratio-Test (LRT) for Model Comparison (REML-estimator)

Name   |   Model |  df | df_diff |   Chi2 |      p
-------------------------------------------------
model2 | lmerMod | 222 |         |        |
model3 | lmerMod | 223 |       1 | 184.28 | < .001
```

Generalized model fitting (negative binomial distribution)

```
anova(model3)
```

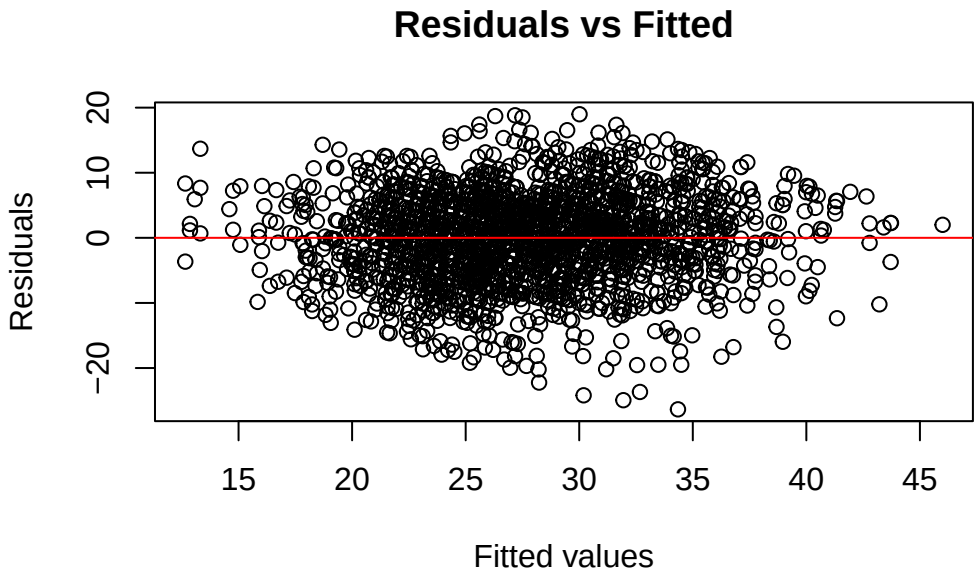
```
Analysis of Variance Table
     npar Sum Sq Mean Sq F value
ivip  219  22146  101.12  1.9492
```

```
#variance stability

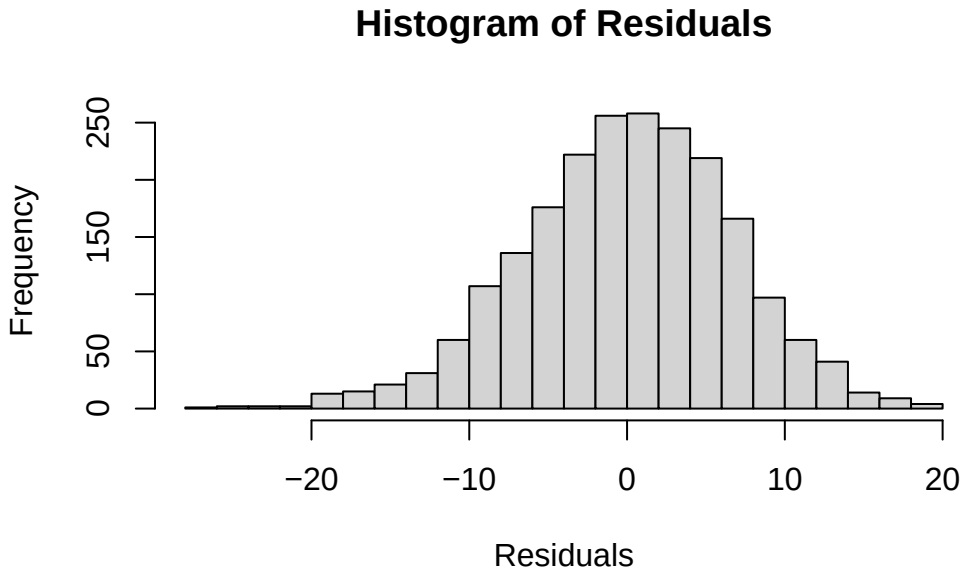



# Extract fitted values and residuals
fitted_values <- fitted(model3)
residuals_values <- residuals(model3)

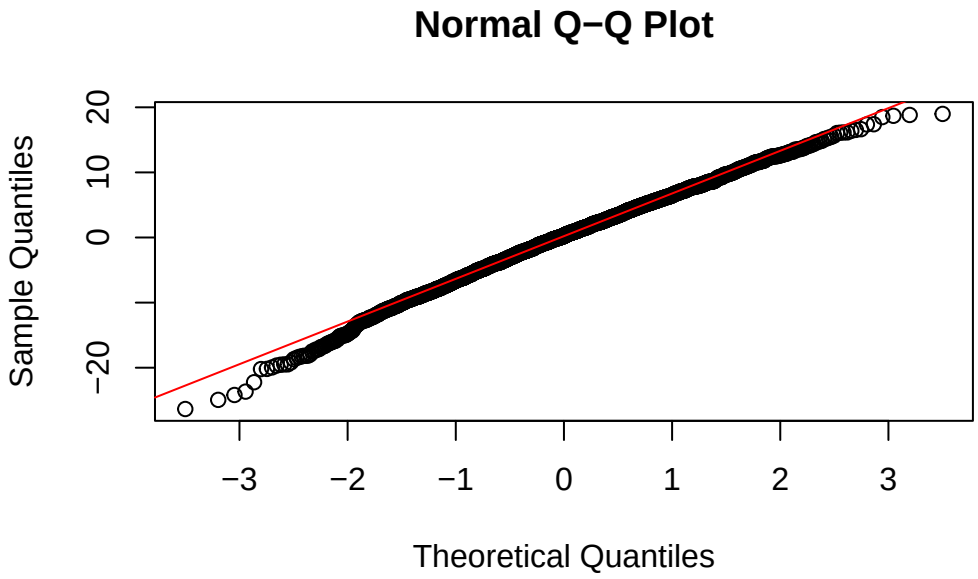
#homosedasticity
# Plot residuals vs fitted values
plot(fitted_values, residuals_values,
     main="Residuals vs Fitted",
     xlab="Fitted values",
     ylab="Residuals")
abline(h=0, col="red")
```



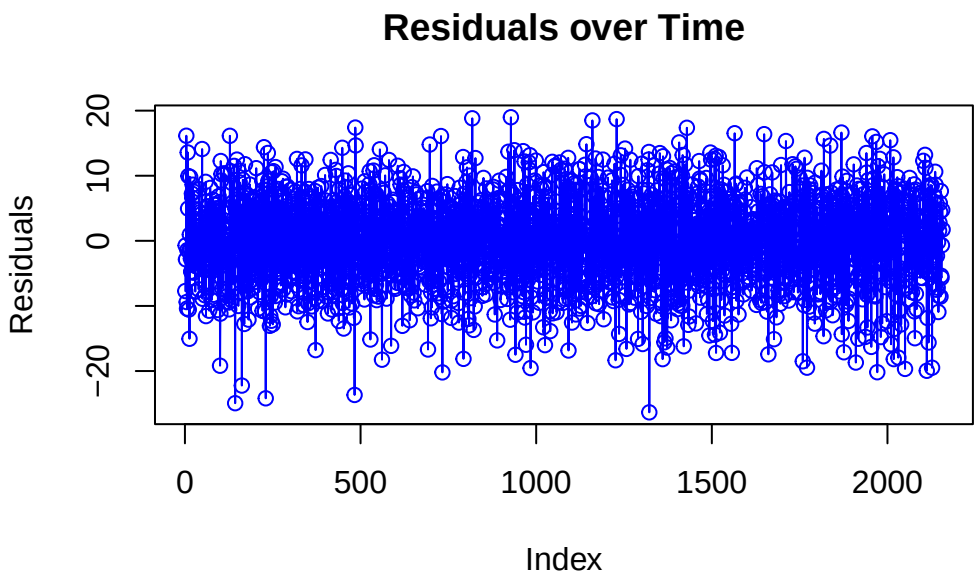

```r
#residual normality
hist(residuals_values, main="Histogram of Residuals", xlab="Residuals", breaks=20)
```



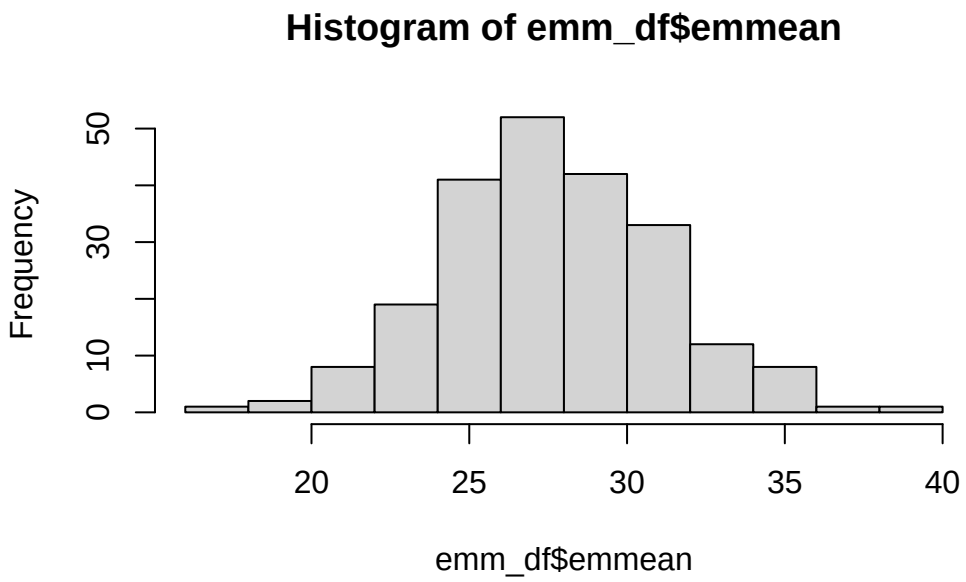
```r
# 2. Q-Q Plot for Residuals
qqnorm(residuals_values)
qqline(residuals_values, col = "red")
```




```r
#residual independance
# Plot residuals over time (for time-series data)
plot(residuals(model3), type="o", col="blue", main="Residuals over Time", xlab="Index", ylab=
```




```
# Extract EMMeans for genotype
emm_genotype <- emmeans(model3, ~ ivip)
emm_df <- as.data.frame(emm_genotype)
hist(emm_df$emmean)
```


Histogram of emm_df$emmean
Frequency
0
10
30
50
20
25
30
35
40
emm_df$emmean